Departamento de Engenharia Mecânica
Área Científica de Mecânica Aplicada e Aeroespacial

Aircraft Optimal Design

MEAer

4^{th} Assignment (Final Project) — 2^{nd} semester – 2025/26

Report due date: 28 June — Handout date: 1 June

Instructions: Install `OpenMDAO` and `OpenAeroStruct` following the instructions included in the appendix of the class notes. This is a prerequisite to complete this assignment. Before the due date, email a brief *report* (**max. 24 pages**) with the results and the *source code* that reproduces the solutions. Add your student numbers as filenames prefix (e.g. report_123456_234567_345678_456789.pdf and code_123456_234567_345678_456789.zip).

Final Project

(20 marks)

The goal of this project is to conduct a conceptual aircraft aerostructural design using `OpenAeroStruct` as the numerical optimization tool.

You should identify an existing aircraft (small, medium or large; manned or unmanned) to be used as reference. Obtain from references relevant parameters such as those in Tab.1.

Specifications	
Range	Cruise Mach number
Cruise altitude	Wing span
Wing sweep (@c/4)	Wing taper ratio
Wing aspect ratio	Thickness/chord ratio
Max. section lift coefficient	Engine SFC
Take-off weight	Operating empty weight
Structural material	Load factor

Table 1: Some relevant aircraft specifications.

The design should include both main wing and horizontal stabilizer. Constraints should include cruise trim in both vertical force and pitching moment. In addition, use multi-point optimization to add a maneuver condition to size the wing spar considering a selected load factor n.

As an aircraft designer, you have total freedom on how to setup your problem...

The report should cover at least the following aspects:

- Selection of aircraft type and mission definition;
- List of aircraft baseline parameters and other data used;
- Formulation of the optimization problem in standard form;
- Discussion of the choice of design variables, constraints and objective;
- Identification of the coupling between disciplines and corresponding connections among modules (N^2 diagram);
- Characterization of the baseline (initial) solution;
- Evolution in complexity from a few to the complete set of design variables;
- Discussion of the optimal solution(s) found;
- Analysis of the optimization history and computational cost;
- Identification of model short falls to handle the design problem (if any);
- List of references used.

Notice, the saying "a picture is worth a thousand words" should not apply in the report. A discussion should follow after every table or figure.

Bonus marks

Additional marks are given to code developments made, e.g., new objective function, constraint or design variable.